Swagato Barman Roy | Résumé

49A | Montara Tower 2B | 1 Lohas Park Road | Tseung Kwan O, Hong Kong
+65 94668329 (UTC+8h) • swagatopablo@aol.com
sg.linkedin.com/in/barmanroys https://github.com/barmanroys

Data and Machine learning architect with a PhD in Computer Engineering, delivered multiple commercial platforms following scalable microservice architecture. Open to opportunities to lead a team using my data development and MLOps skills. Strong preference for working with open source frameworks using clean code / architecture principles

Personal Information

Date of Birth: March 21, 1988 **Citizenship**:Indian **Hong Kong Status**: General Employment Visa

Experience

- **Hong Kong Shanghai Banking Corporation** — **Hong Kong**
  Financial Data Developer, *May 2025-Present*
- **Bullish Hong Kong Limited** — **Central, Hong Kong**
  Data Platform AI Engineer, *November 2024–February 2025*
- **Comfort Delgro Transportation Private Limited** — **Bishan, Singapore**
  Machine Learning Engineer, *November 2023–October 2024*
- **Agency for Science, Technology and Research** — **Jurong West**
  Machine Learning Researcher, *March 2021–September 2023*

Academic Qualifications

- **Nanyang Technological University** — **Jurong West, Singapore**
  Ph.D School of Computer Engineering, Coursework grade: 4.5/5 *2012–2017*
- **Indian Institute of Technology Bombay** — **Mumbai, India**
  Master of Technology in Communication & Signal Processing, Grade: 9/10 *2010–2012*

Abilities

- **Languages:** Proficient (including asynchronous and concurrent programming) in: *C/C++*, *Python*, *Bash*, *SQL*, learning *Rust*, experience implementing object oriented principles and SOLID architecture
- **Big Data and Computing Frameworks:** Databricks, Apache Spark, ORM with SQL-Alchemy and Alembic, NoSQL databases (REDIS, MongoDB), Streaming with Kafka, RabbitMQ, ELT pipelines with DBT for Petabyte scale data, Iceberg and Trino for ACID compliant lakehouse, Microsoft Fabric for analytics
- **Machine Learning:** Supervised learning, Tensorflow, PyTorch, Gluon, time series forecasting with GluonTS and DART, NLP with BERT, experienced developing and managing end to end machine learning systems
- **Gen AI Exprience:** GenAI services from Vertex, Anthropic, LLM Fine tuning with HuggingFace, Multi-agent experience and workflow design with SmolAgents
- **DevOps Skill:** Container orchestration with Kubernetes. GitLab for continuous integration, Keel, Helm and ArgoCD for automated deployments.
- **DevOps and Cloud Experience:** Leveraging AWS, Azure and Oracle cloud for scaleable application and pipeline deployment using Kubernetes, learning Terraform
- **General Skill**: Interpersonal and communication skills, verbally and through technical writings, leadership skill to manage projects, bring clear process and action items based on ambiguous requirements/specifications
- **Academic Strengths:** Published papers on Federated learning, reinforcement learning, generative transformer models
- **Finance:** Familiarity with options pricing, asset classes, blockchain technology, Black-Scholes model and libraries such as quantlib and pyfolio, hands on understanding of FIX protocol, data feed (via quandl), experience trading in OANDA via automated APIs, Bloomberg xbbg and blpapi libraries, Refinitiv (Reuters) data library for python, Orderbook management via Binance APIs, personal trading via Quantconnect and Lean

Selected Projects

- Led a team on slippage modelling in Trade Execution for Crypto Trader Clients
  - Problem formulation based on past order book data, by defining three bands of slippage-low, medium, high
  - Collection and preprocessing of data using past order-book samples and Blockchain mempool data
  - Feature preprocessing and model training involving combination of polars, autogluon, darts and sklearn for time

series and classification
  - Deployment on K8s cluster with ray and Kubernetes
  - Outcome: 80% F1 score on classification model, about 20% jump on trading volume and reduction on slippage measured by VWAP
- **Personal Project**: Sentiment driven trading strategy deployment on Quant-connect (with Python)
  - Formulation of a simple sentiment driven trading strategy (using Finbert model)
  - Integration of text feed and sentiment measurement APIs from Twitter, news APIs
  - Simulated trading to optimise parameters
  - Live trading (with nominal amounts) to gauge Alpha and Beta
  - Tools and Libraries: Twitter feed, beautifulsoup, AWS S3 for data ingestion, FinBert for sentiment analysis, polars for tabular data manipulation, QCAlgorithm for trading with a small portfolio
- In-house RAG and Agentic System Deployment
  - Retrieval augmented generation based on in-house documents, answering client questions of FIX and connectivity protocols for trading
  - Multimodal capability with VertexAI (experimenting with Qwen open source model)
  - Agentic captability for summarisation, database query (chat with tabular data), slack and email integration with LlamaIndex tools and Crew AI
  - Current enhancement on incorporating Model Context Protocol with Anthropic as the standard for agentic communication
  - Outcome: More than 70% time and resource saving of interdeparmental stakeholders
- Led a Team for Audience Clustering of Users with Undisclosed Data and Building End-to-End Pipeline from Clickstream Data Processing to BI Dashboard
  - Frameworks used for data engineering: Spark, Airflow
  - Major machine learning frameworks used: XGBoost, Scikit, Tensorflow
  - Established Model-as-a-Service (MaaS) pipeline (catering up to a million users every hour) and CI/CD for model refresh. Technology stack used: Redis (message broker), Celery (parallel task processing), Uvicorn (for gateway server), Docker (for cloud native deployment).
  - Resulted in up-to 12% improvement in advertisement targeting and click rate.
- Improve Content Recommendation System to generate 15% more user engagement
  - Modelling user's clickstream data using content attributes and similarity across multiple pageviews
  - Model serving via Kubernetes cluster using combination of ClusterIP and Node-Balancing Services for scalability (handling up to 10,000 requests a second)
  - Major frameworks used: PySpark for data engineering, and scikit-learn, FAISS as a similarity search engine, catboost and tf-agents for modelling, MongoDB for document database
- Credit Risk Analytics Platform (HSBC Global Treasuries Financial Engineering Team)
  - Led a team of three developers to integrate frontend, backend along with HSBC's proprietary database tool (Grade) and Model repository (Zeta)
  - Implemented transaction and asynchronous processing using tools cutting down manual workflow by more than 60% to invoke models such as PD, LGD, and EAD
  - Built an extensible model monitoring and triggering system using SOLID, testable architecture
  - Tools used: Airflow, Spark, Celery (for data processing), Uvicorn (for backend), S3 API (for datalake), Tensorflow and Gluon (for modelling purpose)
- **Community Service:** A*STAR software engineering community-of-practice delivering tutorials on cloud architecture best practices.

Awards

- Best paper finalist award in IEEE Symposium of Industrial Electronics & Applications, 2022.
- Best paper award in UAS: Opportunities and Challenges track as part of IEEE ICNS, Washington DC, April 2017.

Selected Publications

- 2023. "Multi-hop Computational Offloading with Reinforcement Learning for Industrial IoT Networks." In IEEE Vehicular Technology Conference, Florence.
- 2022. "Spatio-Temporal Attention with Symmetric Kernels for Multivariate Time Series Forecasting." In IEEE Conference on Industrial Electronics and Applications, Chengdu.
- 2017. "Subspace Based Interference Cancellation in Cooperative Multicell Transmission." In IEEE Vehicular Technology Conference Spring, Sydney.